

Revised Generation Interconnection System Impact Study Report

For Queue Project Q19-1223

Aes Warrior Run Cogeneration Facility 138 kV
June 23, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1223. The Interconnection Customer proposes a 65.8 MW battery energy storage facility interconnecting at the Aes Warrior Run Cogeneration Facility 138 kV substation in the Maryland area. The request was received on June 09, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 300 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 11 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$14,105,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Nipetown - Dam No. 5 Hydro Station 138 KV L
Contingency: American - Urban 345 KV
Loading: 117.0% (Rating: 314 MVA)
DFAX: 10.1%
Pre-project loading: 114.8%
2. Facility: Catawba Island Substation - Tengen 138 KV line
Contingency: Urban - American 345 KV
Loading: 111.8% (Rating: 256 MVA)
DFAX: 10.9%
Pre-project loading: 109.0%
3. Facility: Lascassas Substation - Dam No. 5 Hydro Sta Substation 138 KV line
Contingency: American - Urban 345 KV Line
Loading: 110.2% (Rating: 347 MVA)
DFAX: 10.1%
Pre-project loading: 108.3%
4. Facility: Schafrath Switch - Dam No. 4 Hydro Station 138 KV
Contingency: American - Urban 345 KV
Loading: 108.5% (Rating: 317 MVA)
DFAX: 10.1%
Pre-project loading: 106.4%
5. Facility: Catawba Island - Offerman 138 KV L
Contingency: American - Urban 345 KV
Loading: 107.8% (Rating: 254 MVA)
DFAX: 10.9%
Pre-project loading: 105.0%

6. Facility: Anita - Offerman 138 KV
Contingency: American - Urban 345 KV Line
Loading: 106.3% (Rating: 245 MVA)
DFAX: 10.9%
Pre-project loading: 103.3%
7. Facility: Antelope Mine - Aes Warrior Run Cogeneration Facility 138 KV
Contingency: American - Urban 345 KV Line
Loading: 103.8% (Rating: 300 MVA)
DFAX: 10.9%
Pre-project loading: 101.4%
8. Facility: Anita - Sintek 138 KV
Contingency: American - Urban 345 KV
Loading: 101.9% (Rating: 241 MVA)
DFAX: 10.9%
Pre-project loading: 99.0%
9. Facility: Nipetown - Schafrath Switch 138 KV L
Contingency: American - Urban 345 KV
Loading: 101.1% (Rating: 354 MVA)
DFAX: 10.1%
Pre-project loading: 99.2%
10. Facility: Rio Dell - St. John 138 KV
Contingency: American - Urban 345 KV
Loading: 100.5% (Rating: 513 MVA)
DFAX: 5.9%
Pre-project loading: 99.7%
11. Facility: Plant 31 Paper Mill - Antelope Mine 138 KV line
Contingency: American - Urban Substation 345 KV
Loading: 100.4% (Rating: 308 MVA)
DFAX: 10.9%
Pre-project loading: 98.1%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Nipetown - Dam No. 5 Hydro Station 138 KV L	Rebuild line with higher-capacity conductor	\$2,558,000	42% (\$1,061,000)
Catawba Island - Tenen 138 KV	Rebuild line with higher-capacity conductor	\$9,539,000	60% (\$5,743,000)
Lascassas Substation - Dam No. 5 Hydro Sta Substation 138 KV line	Rebuild line with higher-capacity conductor	\$3,201,000	42% (\$1,327,000)
Schafrath Switch - Dam No. 4 Hydro Station 138 KV L	Reconductor line	\$651,000	42% (\$270,000)
Catawba Island - Offerman 138 KV L	Reconductor line	\$1,467,000	60% (\$884,000)
Offerman - Anita 138 KV	Reconductor line	\$1,700,000	60% (\$1,023,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Antelope Mine - Aes Warrior Run Cogeneration Facility 138 KV	Reconductor line	\$1,696,000	60% (\$1,021,000)
Anita - Sintek 138 KV	Reconductor line	\$849,000	60% (\$511,000)
Nipetown - Schafrath Switch 138 KV Line	Reconductor line	\$639,000	42% (\$265,000)
Rio Dell - St. John 138 KV	Reconductor line	\$2,537,000	12% (\$294,000)
Plant 31 Paper Mill - Antelope Mine 138 KV line	Reconductor line	\$2,834,000	60% (\$1,706,000)
	Total allocated to project		\$14,105,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.